

Explainable Machine Learning Framework for Chronic Kidney Disease Prediction Using SHAP-Based Interpretability Analysis

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Abstract—Artificial Intelligence (AI) and Machine Learning (ML) are increasingly being adopted in healthcare systems for disease prediction, clinical risk assessment, and decision-support applications. Although machine learning models can achieve strong predictive performance, many healthcare AI systems continue to operate as black-box models that provide limited insight into how predictions are generated. In clinical environments, predictive accuracy alone is insufficient, as healthcare professionals require transparent and interpretable explanations to support trust and informed decision-making.

Chronic Kidney Disease (CKD) is a major global health concern associated with progressive kidney function decline, increased healthcare burden, and elevated mortality risk. Early identification of CKD is essential for timely intervention and improved patient outcomes. Existing CKD prediction studies have largely focused on improving predictive performance, while comparatively less attention has been given to model interpretability and clinical transparency.

This study presents an Explainable Artificial Intelligence framework for Chronic Kidney Disease prediction using a Decision Tree classifier integrated with SHapley Additive Explanations (SHAP). The research utilizes the Chronic Kidney Disease dataset from the UCI Machine Learning Repository, consisting of 400 patient records and multiple clinical attributes related to kidney function assessment.

The proposed framework combines predictive modelling with explainability analysis to provide both accurate disease prediction and transparent interpretation of model behaviour. Experimental results demonstrated strong predictive performance with an accuracy of 1.00 and a cross-validation score of 0.97. SHAP analysis identified clinically significant features including Hemoglobin, Specific Gravity, Blood Urea, Serum Creatinine, and Hypertension as major contributors influencing CKD prediction outcomes.

The findings demonstrate that integrating machine learning with explainability techniques can improve transparency, trustworthiness, and interpretability in healthcare AI systems.

Index Terms—Explainable AI, Machine Learning, SHAP, Chronic Kidney Disease, Healthcare Analytics, Clinical Prediction, Decision Tree, Interpretable Machine Learning

I. INTRODUCTION

Artificial Intelligence and Machine Learning are increasingly being adopted in healthcare systems for disease prediction, clinical risk assessment, and medical decision-support systems. Predictive healthcare models can analyze patient health records and identify patterns associated with disease progression. Chronic Kidney Disease (CKD) has become a

major global healthcare concern due to its increasing prevalence and long-term health complications.

Early detection of CKD is critical because delayed diagnosis may lead to kidney failure, cardiovascular complications, and increased mortality risk. Machine learning approaches can assist healthcare professionals by enabling early-stage prediction using clinical biomarkers and patient health information.

Despite the growing use of machine learning in healthcare, many predictive models operate as black-box systems where the internal reasoning behind predictions remains unclear. In healthcare environments, predictive accuracy alone is insufficient because clinicians must understand why a model generates a particular prediction before integrating it into medical practice.

Explainable Artificial Intelligence (XAI) techniques address this challenge by improving transparency and interpretability in AI systems. SHAP (SHapley Additive Explanations) is one of the most widely used explainability frameworks because it provides both local and global explanations regarding feature contribution and model behavior.

This research focuses on developing an explainable machine learning framework for Chronic Kidney Disease prediction using Decision Tree classification integrated with SHAP interpretability analysis.

A. Research Gap

Machine learning techniques have demonstrated promising results in Chronic Kidney Disease prediction and clinical risk assessment. However, many existing studies primarily emphasize predictive performance while providing limited insight into the reasoning behind model decisions. This lack of interpretability creates challenges for clinical adoption, as healthcare professionals require transparent explanations before integrating AI-assisted predictions into medical decision-making processes.

Furthermore, many predictive models operate as black-box systems that do not clearly identify which clinical features contribute most significantly to disease outcomes. Therefore, there remains a need for explainable machine learning frameworks that combine accurate disease prediction with transparent and clinically meaningful interpretation.

B. Contributions of this Study

The main contributions of this research are as follows:

- Development of an explainable machine learning framework for Chronic Kidney Disease prediction.
- Implementation of a Decision Tree classification model using clinical healthcare data.
- Integration of SHAP-based explainability analysis to provide both global and local model interpretation.
- Identification of clinically important biomarkers influencing CKD prediction outcomes.
- Evaluation of predictive performance using multiple classification metrics and cross-validation.
- Demonstration of how Explainable AI can improve transparency, trust, and interpretability in healthcare decision-support systems.

II. OBJECTIVES

- To analyze the Chronic Kidney Disease healthcare dataset.
- To preprocess and clean clinical healthcare data.
- To implement a Decision Tree classifier for CKD prediction.
- To evaluate model performance using standard classification metrics.
- To apply SHAP explainability techniques for model interpretation.
- To identify clinically important features contributing to CKD prediction.
- To improve transparency and trust in healthcare AI systems.

III. RELATED WORK

Machine learning techniques have been widely adopted for healthcare prediction tasks, particularly for the early detection and diagnosis of chronic diseases. Chronic Kidney Disease prediction has attracted significant research attention due to its increasing prevalence and the importance of early intervention.

Several studies have applied machine learning algorithms such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Naive Bayes, and Artificial Neural Networks to improve CKD prediction performance.

Previous research has demonstrated that machine learning models can effectively identify patterns within clinical and laboratory data for CKD diagnosis. Ensemble-based approaches such as Random Forest and Gradient Boosting have often achieved strong predictive performance because of their ability to capture complex relationships among healthcare variables. However, many of these models operate as black-box systems, making it difficult for healthcare professionals to understand the reasoning behind predictions.

In recent years, Explainable Artificial Intelligence has emerged as an important research area aimed at improving transparency and trust in machine learning systems. Among various explainability techniques, SHAP has gained significant attention due to its strong theoretical foundation and ability to provide both local and global explanations.

Several healthcare studies have successfully applied SHAP to improve interpretability in disease prediction models, including diabetes prediction, cardiovascular disease risk assessment, cancer diagnosis, and kidney disease analysis.

Despite these advancements, relatively few studies have focused on combining interpretable machine learning models with SHAP-based explainability specifically for Chronic Kidney Disease prediction. This study addresses this gap by integrating a Decision Tree classifier with SHAP explainability analysis to provide both accurate prediction performance and clinically interpretable insights.

IV. DATASET DESCRIPTION

The Chronic Kidney Disease dataset used in this study was obtained from the UCI Machine Learning Repository. The dataset contains clinical and laboratory records collected from patients for CKD diagnosis analysis.

A. Dataset Overview

- Total Instances: 400 patients
- Total Features: 25 clinical attributes
- Task Type: Binary Classification
- Target Variable: CKD / Not CKD

B. Important Clinical Features

Important clinical attributes used in this study include:

- Hemoglobin (hemo)
- Specific Gravity (sg)
- Blood Urea (bu)
- Serum Creatinine (sc)
- Packed Cell Volume (pcv)
- Hypertension (htn)
- Red Blood Cell Count (rc)
- Potassium (pot)

Target Variable

- CKD: Chronic Kidney Disease
- Not CKD: No Chronic Kidney Disease

C. Dataset Characteristics

The dataset contains both numerical and categorical clinical attributes commonly used during kidney disease diagnosis and monitoring. Several attributes contain missing values, reflecting real-world healthcare data collection challenges. Therefore, appropriate preprocessing techniques were required before model development.

The dataset is suitable for supervised binary classification tasks where the objective is to predict whether a patient is likely to have Chronic Kidney Disease based on available clinical measurements and laboratory test results. The diversity of clinical variables included in the dataset provides an opportunity to investigate both predictive performance and feature-level interpretability using Explainable AI techniques.

D. Class Distribution

The dataset contains both CKD and non-CKD patient records, providing a balanced benchmark for binary classification tasks. The presence of heterogeneous clinical measurements enables comprehensive evaluation of machine learning models for healthcare prediction while also supporting interpretability analysis through explainable AI techniques.

E. Dataset Limitations

Several important limitations of this dataset must be acknowledged. First, the dataset contains only 400 records, which is relatively small for machine learning tasks and may lead to overfitting, particularly for models with high representational capacity.

Second, all data originates from a single hospital in southern India, limiting geographic and demographic generalizability.

Third, missing values are present across several features, requiring imputation assumptions that may introduce bias.

Fourth, the dataset was collected in 2015 and may not fully reflect current clinical practices or patient demographics.

These limitations are reflected in cautious interpretation of model performance throughout this paper.

V. METHODOLOGY

A. Data Collection

The Chronic Kidney Disease (CKD) dataset was imported into the Python analysis environment using the Pandas library. The dataset contains clinical and laboratory records collected from patients for kidney disease diagnosis and prediction.

B. Data Preprocessing

The preprocessing stage included the following steps:

- Removal of unnecessary identifier columns.
- Conversion of categorical variables into numerical form.
- Missing value handling using SimpleImputer.
- Feature transformation and encoding.
- Train-test splitting for model training and evaluation.

C. Machine Learning Model

A Decision Tree classifier was implemented using the Scikit-learn machine learning library.

The model was trained using:

- Training Dataset: 80
- Testing Dataset: 20

D. Explainability Analysis

SHAP (SHapley Additive Explanations) was employed to interpret model predictions and identify feature-level contributions influencing CKD prediction outcomes.

SHAP analysis enabled:

- Global model interpretation
- Local prediction explanation
- Feature importance analysis
- Transparent healthcare AI interpretation

E. Model Evaluation Strategy

To assess the effectiveness of the proposed framework, multiple evaluation metrics were employed. Accuracy was used to measure overall prediction performance, while Precision and Recall evaluated the model's ability to correctly identify CKD cases and reduce misclassification.

The F1-Score was calculated to provide a balanced assessment of Precision and Recall. Additionally, Receiver Operating Characteristic (ROC) analysis and Area Under the Curve (ROC-AUC) were used to evaluate discriminative capability across classification thresholds.

To assess model stability and generalization performance, k-fold cross-validation was performed. Cross-validation reduces performance estimation bias and provides a more reliable assessment of model robustness.

The overall workflow of the proposed framework consists of data collection, preprocessing, model development, performance evaluation, and SHAP-based explainability analysis.

VI. MACHINE LEARNING IMPLEMENTATION

The Decision Tree classifier was trained using the preprocessed CKD dataset. The model successfully learned relationships between clinical variables and disease outcomes.

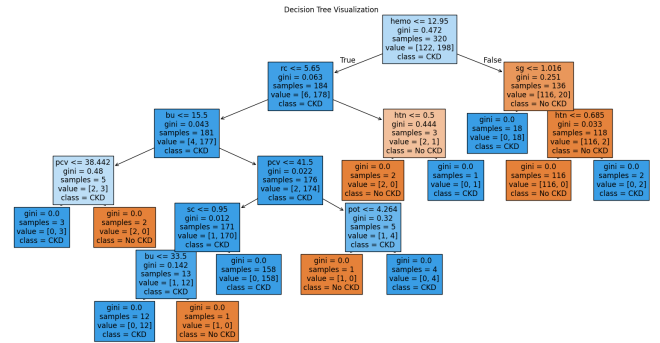


Fig. 1. Decision Tree structure used for CKD prediction

A. Model Selection Rationale

The Decision Tree classifier was selected because of its inherent interpretability and suitability for healthcare applications where transparency is essential. Unlike many black-box machine learning models, Decision Trees provide human-understandable decision paths that can be interpreted by clinicians and healthcare professionals.

Additionally, the model works effectively with mixed healthcare data containing both numerical and categorical variables. The integration of Decision Tree classification with SHAP explainability further enhances transparency by providing feature-level explanations for prediction outcomes.

VII. RESULTS AND PERFORMANCE ANALYSIS

The model was evaluated using multiple classification metrics including Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Cross-Validation.

The proposed Decision Tree model achieved excellent predictive performance across all evaluation metrics. The model successfully distinguished between CKD and non-CKD patients with very high classification accuracy.

Although near-perfect performance was observed, the results should be interpreted cautiously because the dataset size is relatively small and may not fully represent real-world clinical variability. The observed performance may also reflect the relatively clean structure of the dataset and should therefore be validated on larger external clinical datasets before generalizing the findings.

The ROC-AUC score indicates strong discriminative capability, demonstrating that the model can effectively separate positive and negative classes across different classification thresholds. Furthermore, the high cross-validation score suggests that the model maintains stable performance across multiple training and testing splits, indicating good generalization ability.

These results suggest that machine learning can effectively support early-stage CKD prediction using routinely collected clinical and laboratory measurements.

A. Limitations of Performance Evaluation

Although the model achieved excellent performance on the available dataset, the results should be interpreted with caution. The dataset contains only 400 patient records collected from a single source, which may limit the generalizability of the findings.

In addition, the near-perfect evaluation metrics may partially reflect characteristics of the dataset and its relatively limited complexity. Therefore, external validation using larger, more diverse, and multi-center clinical datasets is necessary before drawing conclusions regarding real-world clinical deployment and robustness.

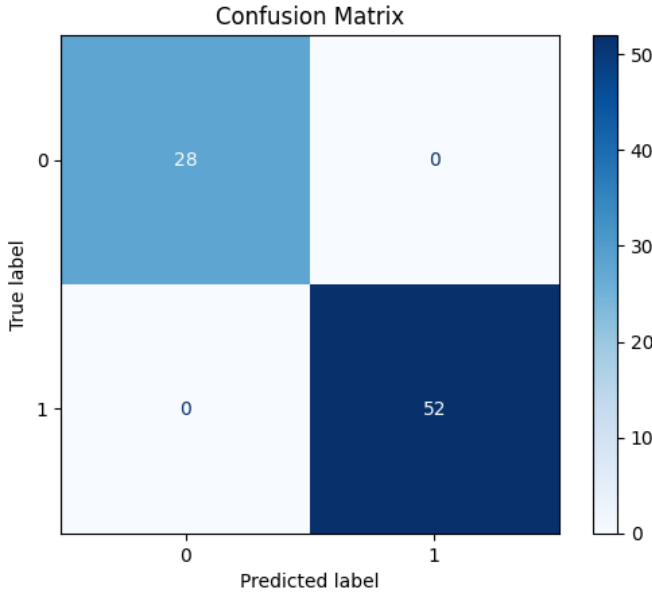


Fig. 2. Confusion Matrix for Decision Tree Classification Model

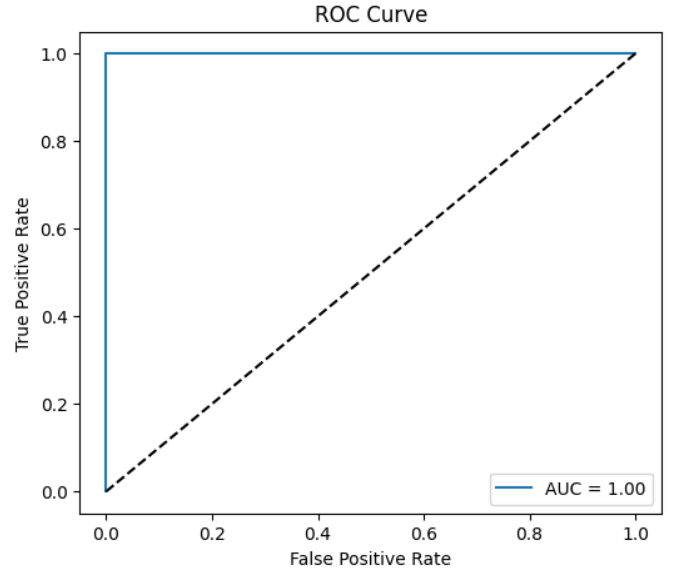


Fig. 3. ROC Curve Showing Classification Performance of the Decision Tree Model

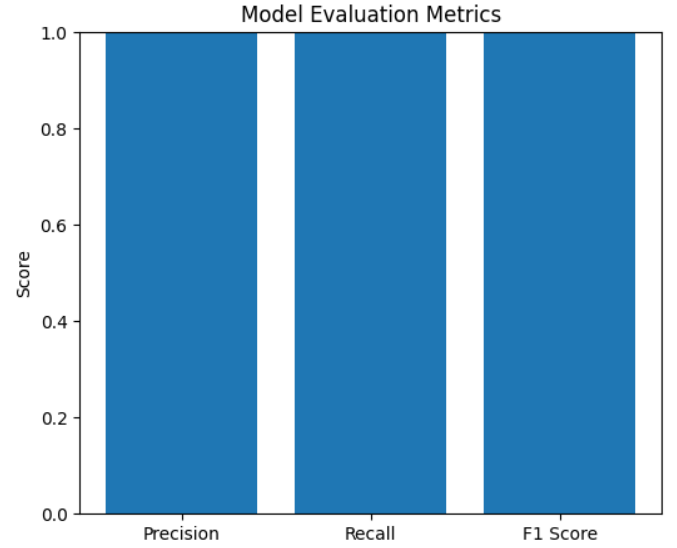


Fig. 4. Evaluation Metrics Comparison for the Proposed Model

VIII. SHAP EXPLAINABILITY ANALYSIS

To improve model transparency and interpretability, SHAP (SHapley Additive Explanations) was applied to the trained Decision Tree model. SHAP is a game-theoretic explainability framework that quantifies the contribution of each feature to a prediction outcome.

The use of SHAP enabled both global and local interpretation of the model. Global explanations identify the most influential features across all predictions, whereas local explanations explain individual prediction outcomes.

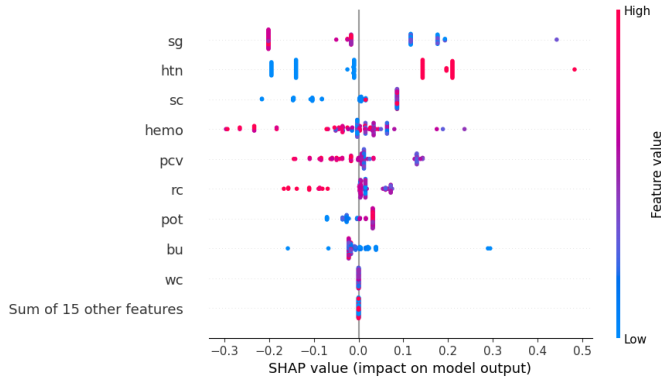


Fig. 5. SHAP Summary Plot Showing Feature Contribution to CKD Prediction

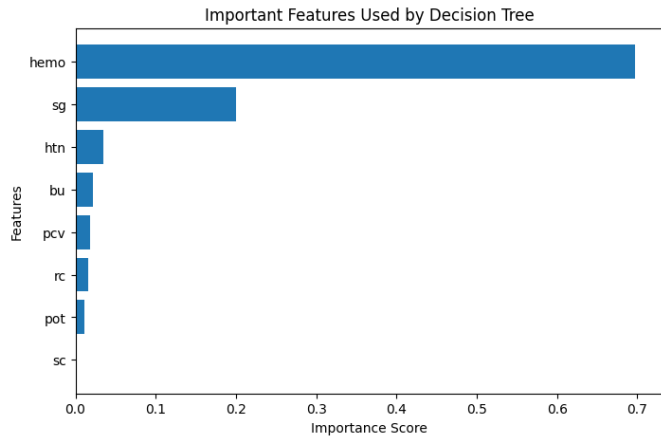


Fig. 6. Global Feature Importance Based on SHAP Values

The SHAP summary plot provides a comprehensive visualization of feature impact on model predictions. Features with larger SHAP values exert greater influence on prediction outcomes. Positive SHAP values increase the likelihood of CKD prediction, whereas negative SHAP values reduce that likelihood.

The global feature importance analysis revealed that Hemoglobin, Specific Gravity, Serum Creatinine, Blood Urea, and Hypertension were among the most influential variables contributing to CKD prediction.

A. Clinical Interpretation of SHAP Results

The SHAP analysis findings are consistent with established clinical knowledge regarding Chronic Kidney Disease diagnosis and monitoring.

Hemoglobin levels are closely associated with anemia, a common complication observed in CKD patients. Similarly, Serum Creatinine and Blood Urea are widely recognized biomarkers used to evaluate kidney function. Specific Gravity reflects the kidney's ability to concentrate urine, while Hypertension is a major risk factor associated with CKD progression.

The ability of SHAP to identify clinically meaningful variables demonstrates that the proposed framework not only

achieves strong predictive performance but also generates medically interpretable explanations that can support healthcare decision-making.

IX. DISCUSSION

This study demonstrates the effectiveness of integrating interpretable machine learning and Explainable Artificial Intelligence techniques for Chronic Kidney Disease prediction. The Decision Tree classifier achieved strong predictive performance while maintaining transparency through SHAP-based explanations.

The findings indicate that explainability techniques can bridge the gap between predictive accuracy and clinical interpretability. By identifying clinically relevant variables influencing prediction outcomes, the proposed framework improves trustworthiness and supports informed healthcare decision-making.

Although the results are encouraging, further validation using larger and more diverse clinical datasets is necessary before practical deployment in healthcare environments.

X. CONCLUSION

This research presented an Explainable Machine Learning Framework for Chronic Kidney Disease prediction using a Decision Tree classifier integrated with SHAP explainability analysis.

Experimental results demonstrated excellent predictive performance, achieving high values across multiple evaluation metrics. SHAP analysis provided transparent insights into feature contributions and identified clinically significant biomarkers associated with CKD prediction.

The integration of machine learning and explainable AI techniques improves transparency, interpretability, and trust in healthcare prediction systems. The proposed framework highlights the potential of Explainable Artificial Intelligence for supporting clinical decision-making and disease risk assessment. The proposed framework demonstrates that explainable machine learning can support transparent and clinically interpretable healthcare prediction systems while maintaining strong predictive performance.

XI. FUTURE WORK

Future research may focus on evaluating the proposed framework using larger multi-center healthcare datasets collected from diverse patient populations.

Additional machine learning algorithms such as Random Forest, XGBoost, LightGBM, and Deep Learning models may also be explored and compared with the Decision Tree classifier.

Further investigation into advanced explainability techniques, counterfactual explanations, and real-time clinical decision-support integration could enhance the practical applicability of explainable healthcare AI systems.

External clinical validation and deployment studies will be important for assessing the robustness and generalizability of the proposed framework in real-world healthcare environments.

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